

Parting Is a Sweet Sorrow

An Ambient Co-Presence Space for Friendships

Drifting Apart in Young Adulthood

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白昕宸 蘇楷峻 顏孜芸

Department of Computer Science and Information Engineering

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Abstract

Adult friendships rarely end in a fight. Far more often they *drift*: as classes split, people relocate, and life paths diverge, the daily texture of incidental contact quietly disappears and a once-close bond fades without any explicit breakup. This pattern — **passive friendship dissolution** — produces a distinctive loneliness that is invisible to the people experiencing it and to the tools meant to help them. We argue that current technologies make the problem worse: social platforms run on a logic of *broadcast, performance, and comparison*, AI companions offer comfort but never the feeling of *being needed*, and friend-matching apps optimize for making *new* ties rather than *maintaining* existing close ones. Across the HCI literature, three design lineages — ambient awareness, digital third places, and asynchronous communication — each solve one half of the problem and leave the same gap open. We propose **Parting Is a Sweet Sorrow**, an ambient co-presence space in which a small group of friends inhabit a shared, projected “night sky” that they build together from privately meaningful objects and a shared vocabulary of light, color, and sound. The system is designed to let members feel the group is *still there* without demanding performance or reply. We pose three research questions — on whether projected shared space sustains felt co-presence (RQ1), on whether non-performative ambient presence lowers comparison and performance pressure while preserving connectedness (RQ2), and on whether the space changes the willingness to maintain a drifting friendship (RQ3) — and outline a mixed-methods evaluation anchored on the Affective Benefits and Costs Questionnaire (ABC-Q).

1 Research Motivation

We frame the motivation with the **ABT** structure (*And, But, Therefore*) required by the project rubric.

1.1 And — friendships are load-bearing infrastructure of young adulthood

Close friendships are not a luxury of young adulthood; they are the developmental scaffolding on which it is built. Emerging adulthood (roughly 18–30) is the life stage of maximal identity exploration and instability [3], and it is also a peak of the lifespan loneliness curve [4, 25]. During exactly this window, friendships do the heavy lifting: they are where young adults build the intimacy skills that underwrite *every* later relationship [26], and the quality of close friendship in adolescence remains a significant long-term predictor of adult emotional health [2]. Mutual close friendships also act as a *buffer*, protecting socially anxious young people against concurrent psychosocial risk [9]. In short, the bonds formed in this period are the foundation for lifelong relational capacity.

1.2 But — friendships dissolve passively, and the drift is invisible

But adult friendships are unusually fragile, and they break in a particular way. *Passive friendship dissolution* is the gradual, unstructured ending of a friendship *without confrontation or any explicit breakup conversation* [33]. The Adult Friendship Dissolution Process Model holds that passive dissolution is steered by a combination of the actor’s personal traits (attachment style, personality), situational pressures (distance, competing life demands), and perceptions of the friend’s likely response — which together push people toward *gradually reducing contact* rather than openly ending the tie [33]. A complementary features → processes → outcomes model of friendship shows that “gradual drift” is a distinct dissolution outcome, separate from direct rejection [28]. Relocation studies confirm the mechanism empirically: most adolescents experience a measurable period of diminished access to companionship and intimacy after a move, with the effect larger for those already socially anxious [32].

The cruelty of this pattern is that it is *invisible*. Because drift causes no acute crisis, it never triggers help-seeking; by the time loneliness arrives, it triggers *withdrawal* rather than outreach [5]. The loneliness itself is a *discrepancy* between the closeness one has and the closeness one wants [10, 12] — not an absence of contact. People in constant digital contact with friends they love can nonetheless feel the bond has quietly downgraded — indeed, young adults themselves report that online contact often lacks the authenticity of in-person interaction and can enable avoidance and curation rather than genuine closeness [29]. And the scale is large: a majority of Gen Z report feeling lonely “sometimes or always” [6], yet “friendship drift” remains essentially unaddressed. Worse, the tools people reach for make it worse: a nine-year longitudinal study found social-media use *predicts increasing* loneliness over time, with the strongest effect among those who use it specifically to maintain close ties [27].

1.3 Therefore — make drift visible and restore ambient co-presence

Therefore we propose to design for this specific loneliness: not the loneliness of having no one, but the loneliness of friendships that have quietly downgraded across diverging paths. Researchers have mapped *who* is at risk and *what* the outcomes look like, but have not yet identified — or designed for — the *daily interaction patterns* that produce passive drift [13, 28]. Our goal is an interactive space that makes drift visible before it becomes irreversible and that restores a sense of *being together* without requiring the performance that today’s tools demand.

1.4 Target audience

Young adults aged roughly 18–30 (“adolescents in transition”) who are passing through the life transitions — graduation, relocation, class- or school-changes, the start of work — that scatter an existing friend group into divergent daily routines. We target groups of close friends (group size ~5) who already share history and want to *maintain* that closeness across distance, rather than strangers seeking to form new ties.

2 Background: How Friendships Work and Why They Drift

To see why passive dissolution is hard to address, we locate it inside a model of how friendship operates. Santucci et al. [28] describe friendship as a chain from *features* (individual behavior and cognition), through *processes* (positive interactions such as companionship and support; negative ones such as conflict), to *outcomes* (maintenance vs. dissolution, where dissolution splits into *gradual drift* and *direct rejection*). Passive dissolution is the gradual-drift branch: it is produced not by any negative interaction but by the slow *subtraction* of positive ones.

This is what makes it hard to design for. The field has mapped the *endpoints* — who is at risk (young adults in transition: relocation, graduation, school change [32, 33]) and what the

final outcome looks like (a friendship observed only after it has dissolved) — but the *middle*, the week-to-week interaction dynamics that tip a bond from “maintained” into “drifted,” remains largely unobserved [28]. **It is hard to design interventions for a process we cannot see.**

Why this loneliness matters — a three-horizon view. The stakes compound over time. **Now**, the gap between attained and desired closeness produces loneliness and depressed mood [10]. **In the mid-term**, the person loses the buffer that good mutual friendships provide against social anxiety [9]. **In the long term**, because adolescent and young-adult friendship quality predicts adult emotional health [2], the damage carries forward. If young adults internalize the lesson that “relationships naturally fade,” they may enter adulthood unwilling to invest in any bond that requires sustained effort — a mass-scale system failure that each individual experiences as personal failure.

3 Related Work

3.1 Loneliness as discrepancy, not absence

A consistent finding across the loneliness literature is that loneliness is a *perceived discrepancy* between desired and actual closeness, not a count of contacts [10, 25]. Emerging adulthood is both the developmental period of greatest relational flux [3] and a peak in the lifespan loneliness curve [4, 25]. Hall et al. [12] characterize young-adult loneliness as a product of “rapid life changes and a lack of relational permanence and routine” rather than of having no one — which is exactly the experiential signature of passive drift. This reframes the design target: the problem is not connecting strangers but *sustaining felt closeness* in existing bonds whose routines have been disrupted.

3.2 Passive vs. active dissolution

Vieth et al. [33] distinguish *active* dissolution (an explicit, often conflict-driven ending) from *passive* dissolution (a gradual withdrawal of contact), and model the actor-, partner-, and context-level predictors that route a friendship toward one or the other. Santucci et al. [28] situate dissolution within the broader friendship lifecycle and identify gradual drift as a distinct outcome with its own dynamics. Recent HCI work corroborates this from the lived-experience side: interviews with people in long-term online friendships find that such ties are sustained through accumulated, low-key interaction and quietly erode when that interaction lapses, suggesting that systems should support *sustained interaction over time* rather than momentary tie-strength [35]. Together these works establish that passive dissolution is a phenomenon in its own right — and that it has been characterized at the level of *predictors* and *outcomes* but not yet at the level of *designable daily interaction*.

3.3 Why current tools fail

Existing technologies fail this problem in four characteristic ways.

The “new friends” paradox.

Friend-matching platforms optimize the *match* but ignore the roughly 200 hours of shared time needed to build a close bond [11]; they manufacture introductions, not maintenance.

Social snacks, not nutrition.

Mainstream social media runs on broadcast, performance, and comparison. It fosters rumination about how one is perceived and unfavorable social comparison [8], and longitudinally *increases* loneliness for the very people using it to stay close [27]. It offers “social snacks” where the need is emotional nutrition.

Table 1: Three design lineages that each solve half of the passive-drift problem.

Lineage		What it contributes	The gap it leaves
Ambient awareness	aware-	Peripheral cues make users aware of another’s ongoing presence <i>without interrupting</i> them, raising felt connectedness [7, 22, 23].	Built almost entirely for <i>one-to-one</i> or family ties (caregiving, couples), not for peer <i>groups</i> drifting apart.
Digital places	third	Persistent shared “places” sustain repeated, low-pressure interaction and casual co-presence [20].	Organized around outward <i>interests</i> and often strangers, not around the shared history of an existing friend group.
Asynchronous communication		Removing the demand for real-time response lowers social anxiety and impression- management load [8, 14].	Treated as a <i>feature</i> of messaging tools, never integrated into the design of a shared interactive <i>space</i> .

Comfort without being needed.

AI companions can provide support and comfort [31], but cannot offer the feeling of *being needed* by another person — the reciprocal interdependence that anchors a genuine close bond.

Breadth over depth.

Interventions and platforms skew toward *making new* ties or treating clinical loneliness, and toward breadth metrics (follower counts, group-chat size). Tools for *maintaining* existing close bonds across distance, and for making drift visible before it is irreversible, are largely missing [1, 13].

3.4 Three HCI lineages, each solving half the problem

The raw materials for a solution already exist in HCI, distributed across three lineages — but each addresses only one half of passive drift, and each leaves the same gap (Table 1).

3.5 The research gap

No existing system unifies all four properties that passive drift demands: *(i)* ambient, peripheral presence that does not interrupt; *(ii)* group-scale rather than dyadic; *(iii)* asynchronous and *non-performative*, removing comparison and reply pressure; and *(iv)* organized around an existing *relationship* and its shared history rather than around interests or strangers.

Recent work has begun to move toward this gap without closing it. Shakeri et al. [30] formalize *passive co-presence* as a design construct — a low-effort, non-demanding sense of another’s presence over distance — but study families rather than peer groups. Leong et al. [21] field-test an ecosystem of distributed, physically-grounded artifacts for staying lightly connected to remote friends, showing real demand for ambient, non-screen-centric connection, yet realize it as discrete *objects* rather than a single shared *space* that a group builds and inhabits together. And Huschke et al. [15] distill design themes specifically for long-distance *friendship* relatedness, confirming that the friendship case — as opposed to romantic or family ties — is a distinct, under-served target. The design space of ambient co-presence has thus been developed productively for transnational families [22] and is only beginning to be charted for friends; no system yet unifies the four properties above for a drifting peer group. We propose to extend *passive co-presence* [30] into exactly this gap.

4 Design Concept: *Parting Is a Sweet Sorrow*

4.1 An ambient co-presence space

We propose a shared, persistent, projected space that a small group of friends *builds together*. The guiding metaphor is a **tent pitched with friends and a shared night sky**. Inside the tent are only things the group mutually understands — a toy everyone once chipped in to buy, a pen someone always used — objects that carry shared memory. Step outside the tent and the sky overhead is full of stars, where *each star is a small spark sent by a friend*. The space is, in one line, “*an ambient atmosphere device, made of the objects that hold your shared memories, that works across distance.*”

Crucially, the canvas is legible *only to the group*. Following the night-sky example: the group shares one sky-canvas; each member can add to it; and what everyone finally sees is the same sky — “*the sky you built together.*” Presence is felt in the periphery, not consumed in the center [34], and is expressed through graspable and ambient media rather than text [17]. The design directly inherits the “presence display” tradition that converts *awareness* into *connectedness* [7, 23].

4.2 Three mechanisms

- M1. A shared interaction vocabulary.** The group agrees on what its signals *mean* — a light or color, a type of object, an ambient sound. Meaning is co-defined and local to the group, not dictated by the platform.
- M2. Private hints.** Signals carry semantics private to the group — e.g., *a purple shimmer might mean “thinking of you.”* An outsider sees only atmosphere; insiders read meaning. This is privacy by *legibility*, not by settings.
- M3. Add / remove objects — trace design.** Members add and remove objects in the space; what accumulates is a *trace* of presence over time, a slow-built record of having-been-here-together rather than a feed of performances.

These mechanisms instantiate the four gap-closing properties: they are ambient (M1, M2), group-scale and relationship-anchored (the shared sky, M3), asynchronous and non-performative (traces accrue without demanding reply), and they encode privacy architecturally through shared-only legibility (M2).

4.3 Positioning against existing systems

Table 2 situates the concept against representative systems in the surrounding design landscape. Each prior system supplies one ingredient; none combines all four.

4.4 Design principles

The concept is governed by four principles drawn from the literature: (1) *Periphery over center* — presence lives at the edge of attention, not as a notification demanding the center [34]; (2) *Ambient and tangible over textual* — meaning is carried by light, color, sound, and objects [17]; (3) *Awareness becomes connectedness* — the design target is felt connectedness, the documented effect of well-made presence displays [7, 23]; and (4) *Non-performative by construction* — because the medium cannot be used to broadcast or compete, it removes the impression-management load that drives both social anxiety [8, 14] and the loneliness paradox of mainstream social media [27]. That such load can be *designed away* is not merely hypothetical: BeReal’s design — randomized timing, reciprocal posting, and no curation — has been shown to suppress self-presentation pressure and reframe social comparison among young adults [19], a precedent our atmosphere-from-objects mechanism extends from momentary snapshots to persistent ambient presence.

Table 2: Design landscape as a property matrix. Columns are the four properties passive drift demands: **Ambient** (peripheral, non-interrupting), **Grp** (group- not dyadic-scale), **A/NP** (asynchronous and non-performative), **Relationship-anchored** (an existing bond, not interests or strangers). ✓ satisfies, ∼ partial, × absent. Each prior system misses at least one; only our system satisfies all four.

Representative system	Amb	Grp	A/NP	Rel	Key limitation for passive drift
Digital Family Portrait [24]	✓	×	✓	✓	One-to-one / caregiving, not a peer group.
Family awareness, e.g. ASTRA [23]	✓	∼	✓	✓	Built and validated for <i>families</i> , not peers.
BuddyBeads [18]	∼	×	✓	✓	Dyadic; fixed small clique, narrow expressivity.
<i>Animal Crossing</i> shared island	×	✓	✓	×	Game/activity-framed, not relationship maintenance.
<i>Between</i> (couples)	×	×	×	✓	Dyadic; creates response-waiting pressure.
<i>Discord</i> friend server	×	✓	∼	×	Performative posting, public comparison, topic-organized.
<i>Parting Is a Sweet Sorrow</i>	✓	✓	✓	✓	Unifies all four for drifting peer groups.

5 Research Objectives and Questions

Objective. To design and evaluate an ambient, non-performative shared space that lets a scattered group of close friends sustain felt co-presence across distance, and to test whether doing so lowers performance and comparison pressure while preserving — or strengthening — the willingness to maintain a drifting friendship.

RQ1 (Foundation).

Can a projected shared space let members *persistently perceive that “the friend group still exists”* without requiring direct, real-time interaction?

RQ2 (Core).

Can this blurry, non-performative co-presence *maintain connectedness while lowering performance and social-comparison pressure*, relative to existing tools?

RQ3 (Real world).

How does living with this space *affect young adults’ willingness and intention to maintain a drifting friendship*, and through what experienced mechanism?

The three questions are deliberately layered: RQ1 tests the *precondition* (felt persistent presence), RQ2 tests the *core claim* (presence without pressure), and RQ3 tests the *downstream relational consequence* in lived use.

6 Evaluation Plan

We propose a mixed-methods evaluation, with one strand per research question (Table 3).

The ABC-Q [16] is central to RQ2 because it was purpose-built to measure the *affective* characteristics of mediated awareness — both its benefits (e.g., connectedness) and its costs (e.g., obligation, pressure) — which is exactly the benefit-without-cost trade-off our design claims to improve. RQ1’s emphasis on *accuracy of perceived presence* rather than message volume operationalizes the calm-technology claim that the system should inform the periphery without capturing the center [34]. RQ3’s qualitative strand is where we expect to surface the

Table 3: Evaluation plan, by research question.

RQ	Approach	Primary measure
RQ1	Test whether the tent atmosphere and shared sky produce persistent awareness that “the group is always there.” We measure the <i>sense of continuous presence</i> , not the <i>number</i> of signals received.	Judgment-accuracy of perceived presence; self-reported continuity of awareness.
RQ2	Compare our space against participants’ existing tools. Hypothesis: <i>affective benefits are maintained while affective costs drop</i> .	Affective Benefits and Costs Questionnaire (ABC-Q): Affective Benefits, Affective Costs, Perceived Awareness [16].
RQ3	Qualitative interviews: does a presence that requires <i>no</i> image-management feel more relaxed, and does it sustain connection? How does this happen?	Thematic analysis of interview transcripts.

experienced mechanism — whether removing performance is what makes sustained, low-effort closeness feel possible.

7 Self-Assessment: McGee’s Eight Questions

8 Expected Contributions

We expect three contributions. *Conceptually*, we name and target a specific, under-served form of loneliness — the loneliness of passive friendship drift — and articulate why mainstream tools structurally worsen it. *In design*, we contribute *Parting Is a Sweet Sorrow*: a group-scale, non-performative ambient co-presence space defined by three mechanisms (shared vocabulary, private hints, trace-based objects) that together close the gap left by prior ambient-awareness, third-place, and asynchronous systems. *Methodologically*, we contribute an evaluation design that separates felt persistent presence (RQ1) from the benefit-without-cost claim (RQ2, via ABC-Q) and from the downstream relational effect (RQ3), giving each a distinct, testable measure.

Parting is such sweet sorrow — the aim of this work is to make the sorrow gentler, and the parting less final, by letting friends remain ambiently, unperformingly, together.

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Table 4: Self-assessment against Prof. Kevin McGee’s eight fundamental questions.

Question	Our answer
What is the research problem?	How to sustain felt co-presence in friend groups undergoing <i>passive</i> dissolution, without the performance and comparison pressure of current tools.
Is the problem significant?	Yes: young-adult loneliness is a peak-vulnerability, high-prevalence phenomenon [4, 6] with lifelong consequences for relational capacity [2, 26].
What is the contribution to knowledge?	A design and an evaluation of <i>group-scale, non-performative ambient co-presence</i> for drifting peer friendships — a gap left by all three existing HCI lineages.
Is the contribution vital?	Drift is invisible and currently un-designed-for; existing tools <i>worsen</i> it [27]. A maintenance-oriented, non-performative alternative is needed.
Originality of the contribution.	Extends the <i>passive co-presence</i> construct [30] and ambient co-presence [22] from families to peer groups, and unifies ambient + group + asynchronous + relationship-anchored design — a combination not present in prior systems (Table 2).
How do readers know it is original?	The landscape and lineage analyses (Tables 1, 2) show each prior system supplies only one ingredient; none combines all four.
Validity of the contribution.	To be established through a layered, falsifiable design (RQ1–RQ3) using a validated instrument (ABC-Q, 16) plus thematic analysis.
Is it appropriate for the discipline?	Yes: it is squarely an HCI / HAX contribution, building on calm technology [34], tangible bits [17], and presence displays [7].

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